

Raphael Ramos

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EDUCATION

University of Alberta

Bachelor of Science in Computer Engineering — EIT (APEGA)

Edmonton, AB

Sept. 2021 – June 2026

TECHNICAL SKILLS

Languages: Python, C, C++, Java, HTML, VHDL, MATLAB (FFT/DSP), ARM, MIPS

Embedded & Hardware: STM32, STM32CubeIDE, Raspberry Pi Pico 2W, Xilinx Zybo Z7 FPGA, Edge Impulse, FreeRTOS, GDB, OpenOCD, JTAG, Vivado, LTSpice, Cadence Virtuoso, WaveForms, KiCad, PCB Design, Fusion 360

Protocols: I2C, SPI, UART, PWM, TCP/UDP sockets, FIFOs

Cloud & Backend: AWS IoT Core, Lambda, DynamoDB, Firebase, MongoDB, MQTT/TLS

Tools & Libraries: Git, GitHub, Make, Docker, Android Studio, VS Code, Linux (CachyOS/Hyprrland, Ubuntu, Debian, Fedora), NumPy, Pandas, Scikit-learn, TensorFlow Lite, OpenMP, pthreads, AI-Assisted Dev (Claude Code), Prompt Engineering

EXPERIENCE

Product Development Intern

May 2024 – Aug. 2024

TURBINE-X Energy Inc.

Nisku, AB

- Performed system-level testing and validation for fully integrated prototypes to ensure specification compliance.
- Calibrated advanced electrical, optical, and sensor equipment ensuring precise measurements across test campaigns.
- Built and brought up servers for internal data sharing, including physical construction and digital configuration.
- Created technical documentation for testing equipment covering training materials, specifications, and test conditions.

Software Deployment Intern

July 2023 – Sept. 2023

Vertical City

Edmonton, AB

- Updated Linux-based software from Ubuntu 18 to Ubuntu 20 across 250+ elevator and lobby screens.
- Troubleshoot LTE connectivity and antenna placement to keep 250+ screens online.
- Re-wired elevator TVs and coordinated deployment with supervisors and building technicians.

PROJECTS

Continuing Care Home Activity Monitor | C++, MicroPython, Edge Impulse, AWS IoT, React Native Jan. – Apr. 2026

- Led the stove-safety subsystem of a 3-subsystem elder-care monitor (stove, water, appliances) and supported work on all three.
- Deployed edge ML on a Raspberry Pi Pico 2W via Edge Impulse, reaching 97.73% accuracy with a sub-10-second response.
- Helped build the AWS IoT pipeline: MQTT/TLS, DynamoDB, Lambda anomaly detection, and Firebase push notifications.
- Helped deliver a caregiver React Native app with real-time alerts and activity history; prototype under \$200 CAD.

UDP Heartbeat Failure Detector | C, VS Code, Make, UDP Sockets, poll() Aug. – Sept. 2026

- Built a liveness monitor in C: agents send periodic UDP heartbeats and a poll()-driven monitor ages silent nodes ALIVE → SUSPECT → DOWN.
- Used CLOCK_MONOTONIC timing and ack-free heartbeats so clock jumps and dropped packets can't trigger false failures.

Multithreaded Scheduler & TCP Object Server | C, pthreads, Mutexes, poll() Mar. 2024

- Built a multithreaded scheduler in C: a pthread per task, mutex-guarded resources, all-or-nothing acquisition with rollback to avoid deadlock, and a monitor thread.
- Built an object-store server: one poll() loop serves 3 concurrent TCP clients over a custom packet protocol; follows a FIFO-based version using select().

RP2040 Motor-Controller Carrier Board | KiCad, PCB Design, Schematic Capture 2026

- Designing a custom 2-layer PCB to replace a breadboard prototype (Pico, H-bridge, pot, LED ring, encoder): schematic capture and footprints in KiCad; layout, ground pour, and JLCPCB fab to follow.

Anomaly Detection System | Python, Scikit-learn, Pandas, Matplotlib Apr. 2025

- Built a detection pipeline using statistical, distance-based (k-NN, DBSCAN), and ML (One-Class SVM) techniques.
- Evaluated models via precision, recall, F1-score, ROC/PR curves; documented end-to-end in a reproducible Jupyter notebook.

16-bit CPU Design | VHDL, Vivado, Xilinx Zybo Z7 FPGA Nov. 2024

- Designed a 16-bit CPU with controller-datapath architecture: ALU, register file, accumulators, and tri-state buffer.
- Developed an FSM executing 13 custom instructions; implemented a bitwise multiplier as a validation test program.
- Resolved double-access load discrepancies through rigorous debugging and simulation in Vivado.

Digital IC Design | Cadence Virtuoso, CMOS, SPICE, WaveForms Fall 2024

- Simulated a complete CMOS digital system in Cadence Virtuoso: inverter, NAND gates, 4-to-1 MUX, TSPC flip-flop, 2-bit shift register.
- Performed transient/DC analysis, transistor sizing, and process-corner/temperature and dynamic-power characterization.

VOLUNTEER

Vice President, Academics

Apr. 2025 – June 2026

Computer Engineering Club, University of Alberta

Edmonton, AB

- Lead academic programming for 100+ members including workshops, study sessions, and exam preparation events.
- Coordinate with faculty and student groups to connect members with academic resources and opportunities.

Ground Station Software Team

Sept. 2023 – Sept. 2024

AlbertaSat, University of Alberta

Edmonton, AB

- Contributed to ground station software development for CubeSat operations and reliable data transmission.
- Collaborated with technical teams on implementation tasks and participated in regular development work sessions.